

# Differential Equation And Linear Algebra.

## 1. Linear Form:

A first-order differential equation is said to be in linear form if it can be written as:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where  $P(x)$  and  $Q(x)$  are functions of  $x$  only.

Eg:-

$$\frac{dy}{dx} + 2y = x$$

Counterexample:

$$\frac{dy}{dx} + y^2 = x$$

## 2. Variable Separable Form:

A differential equation is variable separable if it can be written as:

$$\frac{dy}{dx} = f(x)g(y)$$

$$\text{or } \frac{dy}{g(y)} = f(x) dx$$

so that the variables  $x$  and  $y$  can be separated.

Eg:-

$$\frac{dy}{dx} = xy$$

$$\text{Separating variables: } \frac{dy}{y} = x dx$$

Counterexample:

$$\frac{dy}{dx} = x + y$$

→ First order ordinary differential equation:-

order and degree of a differential equation:-  
order of a DE:-

The order of a differential equation is the highest order derivative present in the equation.

Eg:- ①  $\frac{dy}{dx} + y = x$  (order = 1)

②  $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = 0$  (order = 2)

Degree of a DE:-

The degree of a differential equation is the power of the highest order derivative, provided the equation is free from radicals and fractional powers of derivatives.

Eg:- ①  $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} = 0$  (order = 2, degree = 3)

②  $\left(\frac{dy}{dx}\right)^2 + y = 5$  (order = 1, degree = 2)

①  $e^x dx + e^y dy = 0$   
 → order = 1, degree = 1

②  $\frac{d^2x}{dt^2} + n^2x = 0$   
 → order = 2, degree = 1

③  $y = x \frac{dy}{dx} + \frac{x}{dy/dx}$  → order = 1, degree = 2

④  $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = c$  → order = 2, degree = 2

② Questions:-

- ① Can degree be negative? No  
 ② Can order be a fraction? No.

→ Formation of a Differential Equation:-

① Form the DE of simple harmonic motion given by  $x = A \cos(nt + \alpha)$ . — ①

⇒ diff ① wrt t,

⇒  $\frac{dx}{dt} = A(-\sin(nt + \alpha)n)$   
 $= -An \sin(nt + \alpha)$  — ②

diff ②

$\frac{d^2x}{dt^2} = -An^2 \cos(nt + \alpha)$   
 $= -n^2x$  (from ①)

⇒  $\frac{d^2x}{dt^2} + n^2x = 0$

② Obtain the DE of the system  $x^2 + y^2 + 2ax + c^2 = 0$  where c is a constant and a is a variable.

⇒  $x^2 + y^2 + 2ax + c^2 = 0$

diff wrt x.

⇒  $2x + 2y \frac{dy}{dx} + 2a = 0$

$da = -\left(2x + 2y \frac{dy}{dx}\right)$  — ③

③ In ① ⇒  $x^2 + y^2 - \left(2x + 2y \frac{dy}{dx}\right)x + c^2 = 0$

⇒  $\cancel{\left(2x + 2y \frac{dy}{dx}\right)} \cdot \cancel{x} = \cancel{x^2}$

⇒  $\left(2x + 2y \frac{dy}{dx}\right)x = x^2 + y^2 + c^2$

$$(3) (x-h)^2 + (y-k)^2 = a^2 \quad \text{--- (1)}$$

⇒

diff (1) wrt x

$$\Rightarrow 2(x-h) + 2(y-k) \frac{dy}{dx} = 0$$

$$\Rightarrow (x-h) = -(y-k) \frac{dy}{dx}$$

$$2 + (y-k) \frac{d^2y}{dx^2} + \frac{dy}{dx} \times \frac{dy}{dx}$$

$$(y-k) \frac{d^2y}{dx^2} + \frac{d^2y}{dx^2} = 0$$

$$(y-k) \frac{d^2y}{dx^2} + \frac{d^2y}{dx^2} (y-k) - 1 = 0$$

$$(y-k) = -\frac{d^2y}{dx^2} \times \frac{d^2y}{dx^2}$$

→ Variable Separable Form:

In a differential equation if it is possible to collect all the functions of x and dx on one side and all the functions of y and dy on the other side then the variables are said to be separable. Thus, the general form of such an equation is:

$$f(y) dy = \phi(x) dx.$$

$$(1) \frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$$

$$\Rightarrow \frac{dy}{dx} = e^{-y} (e^x + x^2)$$

$$e^y dy = (e^x + x^2) dx$$

Integrate both sides,

$$\int e^y dy = \int (e^x + x^2) dx$$

$$e^y = e^x + \frac{x^3}{3} + C$$

$$(2) \frac{dy}{dx} + \frac{\sqrt{1+y^2}}{\sqrt{1+x^2}} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\sqrt{1+y^2}}{\sqrt{1+x^2}}$$

$$\frac{1}{\sqrt{1+y^2}} dy = -\frac{1}{\sqrt{1+x^2}} dx$$

Integration on both sides,

$$\sinh^{-1} y = -\sinh^{-1} x + C$$

$$\sinh^{-1} y + \sinh^{-1} x = C$$

$$(3) \frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$$

$$\Rightarrow \frac{dy}{dx} = e^{-2y} (e^{3x} + x^2)$$

$$e^{2y} dy = (e^{3x} + x^2) dx$$

Integrate on both side.

$$\frac{e^{2y}}{2} = \frac{e^{3x}}{3} + \frac{x^3}{3} + \frac{C}{3}$$

$$\Rightarrow e^{2y} = 2 \left( \frac{e^{3x}}{3} + \frac{x^3}{3} + C \right)$$

$$\Rightarrow e^{2y} = \frac{2}{3} (e^{3x} + x^3 + C) \Rightarrow 3e^{2y} = 2(e^{3x} + x^3 + C)$$

$$(4) \frac{dy}{dx} = \frac{x(2 \log x + 1)}{y \sin y + y \cos y}$$

$$\Rightarrow (y \sin y + y \cos y) dy = x(2 \log x + 1) dx$$

Integrate on both sides,

$$\int (y \sin y + y \cos y) dy = \int x(2 \log x + 1) dx$$

$$-y \cos y + y \sin y + \cos y = \int (2x \log x + x) dx$$

$$y \sin y = 2 \int x \log x dx + \int x dx$$

$$y \sin y = 2 \left[ \log x \cdot \frac{x^2}{2} + \frac{x^2}{2} \right]$$

$$\underline{y \sin y = -x^2 \log x + C}$$

$$(1) y - x \frac{dy}{dx} = a \left( y^2 + \frac{dy}{dx} \right)$$

$$\Rightarrow y - ay^2 = \frac{dy}{dx} (a+x)$$

$$y(1-ay) = \frac{dy}{dx} (a+x)$$

$$\frac{dx}{a+x} = \frac{dy}{y(1-ay)}$$

Integrating on both sides,

$$\log|a+x| = \int \frac{dy}{y(1-ay)}$$

$$\log|a+x| = \int \frac{1}{y} dy + \int \frac{a}{1-ay} dy \Rightarrow \frac{1}{y(1-ay)} = \frac{A}{y} + \frac{B}{1-ay}$$

$$\log|a+x| = \log|y| + a \log|1-ay|$$

$$\log|a+x| = \log|y| - \log|1-ay| + C$$

$$(\because \log a - \log b = \log \frac{a}{b})$$

$$\log|a+x| = \log \left| \frac{y}{1-ay} \right| + \log C$$

$$(\because \log a + \log b = \log a \cdot b)$$

$$\Rightarrow \frac{1}{y(1-ay)} = \frac{A}{y} + \frac{B}{1-ay}$$

$$= \frac{A(1-ay) + By}{y(1-ay)}$$

$$\Rightarrow 1 = A - Aay + By$$

$$\underline{1 = A}$$

$$0 = -Aa + B$$

$$0 = -a + B$$

$$\underline{a = B}$$

$$= \frac{1}{y} + \frac{a}{1-ay}$$

$$\log|a+x| = \log C \left| \frac{y}{1-ay} \right|$$

$$\Rightarrow \underline{a+x = C \left( \frac{y}{1-ay} \right)}$$

(2) Find the equation of a curve passing through the point (1,1) whose differential equation is.

$$(y - yx) dx + (x + xy) dy = 0$$

$\Rightarrow$

$$y(1-x) dx + x(1+y) dy = 0$$

Divide by  $xy$ .

$$\left( \frac{1-x}{x} \right) dx + \left( \frac{1+y}{y} \right) dy = 0$$

$$\left( \frac{1}{x} - 1 \right) dx + \left( \frac{1}{y} + 1 \right) dy = 0$$

Integrate, ~~on both~~

$$\log|x| - x + \log|y| + y = C$$

At (1,1),

$$\log 1 - 1 + \log 1 + 1 = C \quad (\because \log 1 = 0)$$

$$\Rightarrow \underline{C = 0}$$

$\therefore$  Particular solution is  $\log|xy| + y - x = 0$

$$(3) (x^2+1) \frac{dy}{dx} + y^2+1=0 ; y(0)=1$$

$$\Rightarrow (x^2+1) \frac{dy}{dx} = -(y^2+1)$$

$$\frac{dy}{y^2+1} = -\frac{dx}{x^2+1}$$

Integrating on both sides,

$$-\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$$

$$-\tan^{-1}y = \tan^{-1}x + C$$

$$\tan^{-1}x + \tan^{-1}y = C$$

$$\tan^{-1}(0) + \tan^{-1}(1) = C$$

$$\boxed{\frac{\pi}{4} = C}$$

$$\Rightarrow \tan^{-1}x + \tan^{-1}y = \frac{\pi}{4}$$

$$\textcircled{1} \frac{dy}{dx} = \sin(x+y) + \cos(x+y) \quad \text{--- ①}$$

$\Rightarrow$  Take

$$x+y = t$$

Diff w.r.t x,

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{dt}{dx} - 1$$

$$\textcircled{2} \Rightarrow \frac{dt}{dx} - 1 = \sin t + \cos t$$

$$\frac{dt}{dx} = \sin t + \cos t + 1$$

$$\Rightarrow \int \frac{dt}{1 + \sin t + \cos t} = \int dx$$

$$\Rightarrow \int \frac{d\theta}{\sin\theta\cos\theta + \cos^2\theta} = x + C \quad (\because t = 2\theta)$$

$$\int \frac{d\theta}{\sin\theta\cos\theta + \cos^2\theta} = x + C$$

$\div$  by  $\cos^2\theta$ .

$$\int \frac{\sec^2\theta d\theta}{\tan\theta + 1} = x + C$$

Take,

$$1 + \tan\theta = u$$

$$\sec^2\theta d\theta = du$$

$$\int \frac{du}{u} = x + C$$

$$\log|u| = x + C$$

$$\log|1 + \tan(\frac{t}{2})| = x + C$$

$$\log|1 + \tan(\frac{x+y}{2})| = x + C$$

$$\textcircled{2} \frac{dy}{dx} = (4x+y+1)^2 - 0, \quad y(0) = 1$$

$\Rightarrow$  Take

$$4x+y+1 = t$$

Diff w.r.t x,

$$4 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{dt}{dx} - 4$$

$\textcircled{3} \Rightarrow$

$$\frac{dt}{dx} - 4 = t^2$$

$$\frac{dt}{dx} = t^2 + 4$$

$$\int \frac{dt}{t^2 + 4} = \int dx \Rightarrow \frac{1}{2} \tan^{-1}\left(\frac{t}{2}\right) = x + C$$

$$\frac{1}{2} \tan^{-1}\left(\frac{t}{2}\right) = x + C$$

$$\Rightarrow \frac{1}{2} \tan^{-1} \frac{1}{2} (4x+y+1) = x+C \quad - \text{eq}$$

$$y(0)=1$$

$$\Rightarrow \frac{1}{2} \tan^{-1} \frac{1}{2} (0+1+1) = 0+C$$

$$\frac{1}{2} \tan^{-1}(1) = C$$

$$C = \frac{\pi}{8}$$

$$\Rightarrow \frac{1}{2} \tan^{-1} \frac{1}{2} (4x+y+1) = x + \frac{\pi}{8}$$

$$\tan^{-1} \frac{1}{2} (4x+y+1) = 2x + \frac{\pi}{4}$$

$$4x+y+1 = 2 \tan(2x + \frac{\pi}{4})$$

→ Exact Differential Equations:-

A first order ODE of the form of  $M(x,y)dx + N(x,y)dy = 0$  — ① is called an exact differential equation.

If the differential form

$Mdx + Ndy$  is exact i.e.,

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

of some function  $u(x,y)$ .

A necessary & sufficient condition for a DE

① to be exact is

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

→ Solution of an Exact DE:-

Solution of eq ① is given as

$$\int_{y=\text{const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\textcircled{1} (y^2 e^{xy^2} + 4x^3) dx + (2xy e^{xy^2} - 3y^2) dy = 0$$

$$\Rightarrow M_y = \frac{\partial M}{\partial y} = y^2 e^{xy^2} \times 2xy + e^{xy^2} \times 2y = 2xy^3 e^{xy^2} + 2y e^{xy^2}$$

$$N_x = \frac{\partial N}{\partial x} = 2y (x e^{xy^2} \times y^2 + e^{xy^2} \times 1) = 2xy^3 e^{xy^2} + 2y e^{xy^2}$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{exact}$$

Sol<sup>n</sup>:-

$$\int_{y=\text{const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\Rightarrow \int (y^2 e^{xy^2} + 4x^3) dx + \int -3y^2 dy = C$$

$$\Rightarrow \frac{y^2 e^{xy^2}}{y^2} + x^4 - y^3 = C$$

$$\Rightarrow e^{xy^2} + x^4 - y^3 = C$$

$$\textcircled{2} \{y(1 + \frac{1}{x}) + \cos y\} dx + \{x + \log x - x \sin y\} dy =$$

$$\Rightarrow \frac{\partial M}{\partial y} = (1 + \frac{1}{x}) - \sin y$$

$$\frac{\partial N}{\partial x} = (1 + \frac{1}{x}) - \sin y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{Exact}$$

$$\text{Sol}^n:- \int_{y \text{ const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\int (y(1 + \frac{1}{x}) + \cos y) dx + \int 0 dy = C$$

$$\Rightarrow y(x + \log x) + x \cos y = C$$

$$\Rightarrow \underline{xy + y \log x + x \cos y = C}$$

$$\textcircled{3} \frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$$

$$\Rightarrow \underbrace{(\sin x + x \cos y + x)}_N dy + \underbrace{(y \cos x + \sin y + y)}_M dx = 0$$

$$\frac{\partial M}{\partial y} = \cos x + \cos y + 1$$

$$\frac{\partial N}{\partial x} = \cos x + \cos y + 1$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{Exact}$$

Sol<sup>n</sup>:-

$$\int_{y \text{ const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\int (y \cos x + \sin y + y) dx + \int 0 dy = C$$

$$\Rightarrow \underline{y \sin x + x \sin y + xy = C}$$

→ DE reducible to exact Form:-

If  $Mdx + Ndy = 0$  is not exact an integrating factor  $\mu(x, y)$  may make it exact.

① If  $\frac{My - Nx}{N} = f(x)$ , then  $\mu = e^{\int f(x) dx}$

② If  $\frac{Nx - My}{M} = g(y)$ , then  $\mu = e^{\int g(y) dy}$

③ If the eq. is homogeneous, then  $\mu = \frac{1}{x^M + y^N}$ , where  $xM + yN \neq 0$ .

$$\textcircled{Q} \textcircled{1} \underbrace{(3xy + y^2)}_M dx + \underbrace{(x^2 + xy)}_N dy = 0 \quad \textcircled{1}$$

$$\Rightarrow \frac{\partial M}{\partial y} = 3x + 2y, \quad \frac{\partial N}{\partial x} = 2x + y$$

$$\frac{My - Nx}{N} = \frac{3x + 2y - 2x - y}{x(x+y)} = \frac{(x+y)}{x(x+y)} = \frac{1}{x}$$

$$\therefore \text{IF} = e^{\int f(x) dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = \underline{x}$$

eq ①  $\times$  IF

$$\Rightarrow \underbrace{(3x^2y + xy^2)}_M dx + \underbrace{(x^3 + x^2y)}_N dy = 0 \quad \textcircled{2}$$

N ( $\because$  multiply with x)

$$\frac{\partial M}{\partial y} = 3x^2 + 2xy, \quad \frac{\partial N}{\partial x} = 3x^2 + 2xy$$

$\therefore$  eq ② is exact

$$\therefore \text{Sol}^n:- \int_{y \text{ const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\Rightarrow \int (3x^2y + xy^2) dx + \int 0 dy = C$$

$$\Rightarrow y \int (3x^2 + xy) dx = C$$

$$\Rightarrow y [x^3 + \frac{xy^2}{2}] = C$$

$$\textcircled{a} (y^4 + xy) dx + (xy^3 + 2y^4 - 4x) dy = 0$$

$$\Rightarrow \frac{\partial M}{\partial y} = 4y^3 + x$$

$$\frac{\partial N}{\partial x} = y^3 - 4$$

$$\frac{\partial N}{\partial y} - \frac{\partial M}{\partial x} = y^3 - 4 - 4y^3 - x$$

$$y^4 + xy$$

$$= -3y^3 - 6 = -3(y^3 + 2)$$

$$y(y^3 + 2)$$

$$\therefore IF = e^{\int (y^3 + 2) dy} = e^{\frac{y^4}{4} + 2y} = e^{\frac{y^4}{4} + 2y}$$

$$= y^{-3}$$

$$(\because a \log b = \log b^a)$$

$$\Rightarrow y^{-3} (y^4 + xy) dx + (xy^3 + 2y^4 - 4x) dy = 0$$

$$\Rightarrow (y + \frac{x}{y}) dx + (x + 2y - \frac{4x}{y}) dy = 0$$

$$\frac{\partial M}{\partial y} = 1 - \frac{4}{y^3}$$

$$\frac{\partial N}{\partial x} = 1 - \frac{4}{y^3}$$

$$\therefore \text{eq. \textcircled{a} is exact.}$$

$$\therefore \text{Sol}^n :- \int M dx + \int N dy = C$$

$$\text{term without } y$$

$$\Rightarrow \int (y + \frac{x}{y^2}) dx + \int 2y dy$$

$$\Rightarrow (y + \frac{x}{y^2}) x + y^2 = C$$

$$\Rightarrow xy + \frac{x^2}{2y^2} + y^2 = C$$

\textcircled{3}

$$(x^2y - 2xy^2) dx + (3x^2y - x^3) dy = 0$$

$$\frac{\partial M}{\partial y} = x^2 + 4xy + x$$

$$\frac{\partial N}{\partial x} = 6xy - 3x^2$$

eq. \textcircled{3} is homogeneous.

$$IF = \frac{1}{xM + yN}$$

$$xM + yN = x(x^2y - 2xy^2) + y(3x^2y - x^3)$$

$$= x^3y - 2x^2y^2 + 3x^2y^2 - x^3y$$

$$= x^3y^2 - 2x^2y^2 + 3x^2y^2 - x^3y$$

$$IF = \frac{1}{x^2y^2}$$

eq. \textcircled{1} \times IF :-

$$\Rightarrow \frac{1}{x^2y^2} (x^2y - 2xy^2) dx + \frac{1}{x^2y^2} (3x^2y - x^3) dy = 0$$

$$\Rightarrow (\frac{1}{y} - \frac{2}{x}) dx + (\frac{3}{y} - \frac{x}{y^2}) dy = 0$$

$$\frac{\partial M}{\partial y} = -\frac{1}{y^2}, \quad \frac{\partial N}{\partial x} = -\frac{1}{y^2}$$

$\therefore$  eq ② is exact.

$\therefore$  Sol<sup>n</sup>:-

$$\int_{y=\text{const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\int \left( \frac{1}{y} - \frac{2}{x} \right) dx + \int \frac{3}{y} dy = C$$

$$\Rightarrow \frac{x}{y} - 2 \log x + 3 \log y = C$$

$$\Rightarrow \frac{x}{y} - \log x^2 + \log y^3 = C$$

$$\Rightarrow \frac{x}{y} + \log \left( \frac{y^3}{x^2} \right) = C \quad \left[ \because \log a - \log b = \log \left( \frac{a}{b} \right) \right]$$

$\rightarrow$  Linear Differential Equation of First Order:-

The general form of linear DE is given as  $\boxed{\frac{dy}{dx} + P(x)y = Q(x)}$  — ①

To solve eq ①, we multiply it with an integrating factor,  $\boxed{IF = e^{\int P(x) dx}}$

$$\therefore \text{eq ① becomes } \Rightarrow e^{\int P(x) dx} y' + e^{\int P(x) dx} P(x)y = e^{\int P(x) dx} Q(x)$$

$$\therefore \text{Sol}^n \text{ of ① } \Rightarrow \boxed{y e^{\int P(x) dx} = \int e^{\int P(x) dx} Q(x) dx + C}$$

$$\textcircled{1} \quad y' + 2y = e^{-x}$$

$$\Rightarrow (y' + P(x)y = Q(x)) \quad \therefore P(x) = 2, \quad Q(x) = e^{-x}$$

$\therefore$  Given eq<sup>n</sup> is linear.

$$\therefore \text{Sol}^n: - y e^{\int P(x) dx} = \int e^{\int P(x) dx} Q(x) dx + C$$

$$\Rightarrow y e^{2x} = \int e^{2x} \cdot e^{-x} dx + C$$

$$\Rightarrow y e^{2x} = \int e^x dx + C$$

$$\Rightarrow y e^{2x} = e^x + C$$

$$\Rightarrow y = e^{-2x} (e^x + C)$$

$$\Rightarrow \underline{y = e^{-x} + C e^{-2x}}$$

$$\boxed{e^{\int P(x) dx} = e^{\int 2 dx} = e^{2x}}$$

$$\textcircled{2} \quad y \log y dx + (x - \log y) dy = 0 \quad \textcircled{1}$$

$$\Rightarrow \frac{dy}{dx} + P(y)y = Q(y)$$

$$\left( \frac{dx}{dy} + P(y)x = Q(y) \right) \rightarrow x e^{\int P(y) dy} = \int e^{\int P(y) dy} Q(y) dy$$

①  $\Rightarrow$  Divide by dy.

$$\Rightarrow (y \log y) \frac{dx}{dy} + (x - \log y) = 0$$

$$\Rightarrow y \log y \left( \frac{dx}{dy} \right) + x = \log y$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{y \log y} = \frac{\log y}{y \log y} = \frac{1}{y}$$

$$\textcircled{2} \text{ is linear, with } P(y) = \frac{1}{y \log y}, \quad Q(y) = \frac{1}{y}$$

$$\therefore IF = e^{\int P(y) dy}$$

$$\int P(y) dy = \int \frac{1}{y \log y} dy \Rightarrow \text{put } \log y = t$$

$$dt = \frac{1}{y} dy$$

$$= \int \frac{1}{t} dt = \log t = \log \log y.$$

$$\therefore IF = e^{\log \log y} = \underline{\log y}$$

$$\therefore \text{Sol}^n: x e^{\log y} = \int e^{\log y} Q(y) dy + C$$

$$\Rightarrow x \log y = \int \frac{\log y}{y} dy + C$$

~~Take  $t = \log y$~~

~~Take  $t = \log y$~~   
 $dt = \frac{1}{y} dy$

$$x \log y = \int t dt + C$$

$$x \log y = \frac{t^2}{2} + C$$

$$x \log y = \frac{(\log y)^2}{2} + C$$

$$x \log y = \frac{1}{2} (\log y)^2 + C$$

$$x \log y - \frac{1}{2} (\log y)^2 = C$$

$$\textcircled{2} \frac{dy}{dx} - y \tan x = \cos x$$

$$\Rightarrow y' + P(x)y = Q(x)$$

$$\therefore P(x) = -\tan x, \quad Q(x) = \cos x$$

$\therefore$  Given eq is linear.

$$e^{\int P(x) dx} = e^{\int -\tan x dx}$$

$$= e^{-\log |\sec x|}$$

$$= e^{\log \sec x} = e^{\log \frac{1}{\cos x}}$$

$$= e^{\log \cos x} = \cos x$$

$\therefore$  Sol<sup>n</sup>:-

$$y e^{\int P(x) dx} = \int e^{\int P(x) dx} Q(x) dx + C$$

$$y \cos x = \int \cos^2 x dx + C$$

$$y \cos x = \frac{1}{2} \int (\cos 2x + 1) dx + C$$

$$\cos 2x = \cos^2 x - \sin^2 x$$
  
 $\cos 2x = 2\cos^2 x - 1$   
 $\Rightarrow \cos^2 x = \frac{\cos 2x + 1}{2}$

$$y \cos x = \frac{1}{2} \int (\cos 2x + 1) dx + C$$

$$y \cos x = \frac{1}{2} \left[ \frac{1}{2} \sin 2x + x + C \right]$$

$$y \cos x = \frac{1}{4} \sin 2x + \frac{x}{2} + C$$

$$y \cos x = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

$$y = \frac{1}{\cos x} \left( \frac{x}{2} + \frac{\sin 2x}{4} + C \right)$$

$$y = \sec x \left( \frac{x}{2} + \frac{\sin 2x}{4} + C \right)$$

$$\Rightarrow y = \frac{x \sec x}{2} + \frac{\sin 2x \sec x}{4} + C \sec x$$
  

$$\Rightarrow y = \frac{x \sec x}{2} + \frac{\sin x}{2} + C \sec x$$

$$\textcircled{4} (x+1) \frac{dy}{dx} + y = (x+1)^3 e^x$$

$\Rightarrow$  Divide by  $(x+1)$ ,

$$\frac{dy}{dx} + \frac{y}{x+1} = (x+1)^2 e^x$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$\therefore$  Given eq is linear.

$$P(x) = \frac{1}{x+1}, \quad Q(x) = (x+1)^2 e^x$$

$$IF = e^{\int P(x) dx} = e^{\int \frac{1}{x+1} dx}$$

$$= e^{\log(x+1)}$$

$$\therefore \text{Sol}^n: IF = x+1$$

$$y e^{\int P(x) dx} = \int e^{\int P(x) dx} Q(x) dx + C$$

$$y(x+1) = \int (x+1)(x+1)^2 e^x dx + C$$

$$(x+1)y = \int (x+1)^3 e^x dx + C$$

$$(x+1)y = \int (x+1)^3 e^x dx + C$$

(product rule is done)

~~$$(x+1)y = e^x [(x+1)^3 - 3(x+1)^2 + 6(x+1) - 6]$$~~

$$(x+1)y = e^x [(x+1)^3 - 3(x+1)^2 + 6(x+1) - 6]$$

$$= e^x [x^3 + 3x^2 + 3x + 1 - 3x^2 - 6x - 3 + 6x + 6 - 6]$$

$$(x+1)y = e^x [x^3 + 3x - 2]$$

$$y = \frac{e^x (x^3 + 3x - 2) + C}{x+1}$$

$$(5) (x+1) \frac{dy}{dx} - y = e^{3x} (x+1)^2$$

$\Rightarrow$  Divide by  $(x+1)$ ,

$$\frac{dy}{dx} - \frac{y}{x+1} = e^{3x} (x+1)$$

$\therefore$  given eq is linear,  $\frac{dy}{dx} + P(x)y = Q(x)$   
 $\therefore P(x) = -\frac{1}{x+1}$ ,  $Q(x) = e^{3x} (x+1)$

$$IF = e^{\int P(x) dx}$$

$$= e^{\int -\frac{1}{x+1} dx}$$

$$= e^{-\log(x+1)} = \frac{1}{x+1}$$

$\therefore$  Sol<sup>n</sup>: -

$$y e^{\int P(x) dx} = \int e^{\int P(x) dx} Q(x) dx + C$$

$$\frac{y}{x+1} = \int \frac{1}{x+1} [e^{3x} (x+1)] dx + C$$

$$\frac{y}{x+1} = \int e^{3x} dx + C$$

$$\frac{y}{x+1} = \frac{e^{3x}}{3} + C \Rightarrow y = (x+1) \left( \frac{e^{3x}}{3} + C \right)$$

$$(6) \cos^2 x \frac{dy}{dx} + y = \tan x$$

$\Rightarrow$  Divide by  $\cos^2 x$ ,

$$\frac{dy}{dx} + \frac{y}{\cos^2 x} = \frac{\tan x}{\cos^2 x} \quad \left( \because \frac{1}{\cos^2 x} = \sec^2 x \right)$$

$$\frac{dy}{dx} + y \sec^2 x = \tan x \sec^2 x$$

$\therefore$  eq is linear,

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$P(x) = \sec^2 x, \quad Q(x) = \tan x \sec^2 x$$

$$\therefore \text{Sol}^n: -$$

$$IF = e^{\int P(x) dx} = e^{\int \sec^2 x dx} = e^{\tan x} = \underline{\underline{e^{\tan x}}}$$

$$y e^{\int P(x) dx} = \int e^{\int P(x) dx} Q(x) dx + C$$

$$y e^{\tan x} = \int e^{\tan x} \tan x \sec^2 x dx + C$$

Take,  $t = \tan x$

$$dt = \sec^2 x dx$$

$$y e^{\tan x} = \int t e^t dt + C$$

$$y e^{\tan x} = e^t (t-1) + C$$

$$y = \frac{e^{\tan x} (\tan x - 1) + C}{e^{\tan x}}$$

$$y = \tan x - 1 + C e^{-\tan x}$$

⇒ Bernoulli's ~~Principle~~ Differential Equation:-

The equations of the form  $\frac{dy}{dx} + P(x)y = Q(x)y^n$ , where  $n \neq 0, 1$  are called Bernoulli's differential equation.

\* Solution methodology:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \text{--- ①}, \quad n \neq 0, 1$$

÷ ① with  $y^{-n}$ ,

$$\Rightarrow y^{-n} \frac{dy}{dx} + y^{1-n} P(x) = Q(x) \quad \text{--- ②}$$

Substitute  $y^{1-n} = z$ ,

$$(1-n)y^{1-n-1} \frac{dy}{dx} = \frac{dz}{dx}$$

$$(1-n)y^{-n} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\therefore y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dz}{dx} \quad (\text{Divide by } (1-n))$$

$$\text{②} \Rightarrow \frac{1}{1-n} \frac{dz}{dx} + zP(x) = Q(x)$$

$$\Rightarrow \frac{dz}{dx} + (1-n)zP(x) = (1-n)Q(x) \quad \text{--- ③}$$

∴ ③ is linear.

$$\text{①} \frac{dy}{dx} + y = xy^2 \quad \text{--- ①}$$

⇒ Here,  $P(x) = 1$ ,  $Q(x) = x$

Divide ① with  $y^2$ ,

$$\Rightarrow y^{-2} \frac{dy}{dx} + y^{-1} = x \quad \text{--- ②}$$

$$\text{Substitute } y^{-1} = z,$$

$$-1 y^{-2} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\therefore -y^{-2} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\text{②} \Rightarrow -\frac{dz}{dx} + z = x$$

$$\Rightarrow \frac{dz}{dx} - z = -x \quad \text{--- ③}$$

∴ ③ is linear,

$$P(x) = -1, \quad Q(x) = -x$$

$$\text{IF} = e^{\int P(x)dx} = e^{\int -1dx} = e^{-x}$$

$$\therefore z e^{\int P(x)dx} = \int e^{\int P(x)dx} \cdot Q(x) dx + C$$

$$z e^{-x} = \int e^{-x} \cdot (-x) dx + C$$

$$z e^{-x} = -\int e^{-x} x dx + C \quad \left[ \int u dv = u \int dv - \int \left( \frac{du}{dx} \int dv \right) dx \right]$$

$$z e^{-x} = -\left[ x \int e^{-x} dx - \int \left[ \int e^{-x} dx \right] dx \right]$$

$$z e^{-x} = -\left[ -x e^{-x} - \int -e^{-x} dx \right]$$

$$z e^{-x} = -\left[ -x e^{-x} + (-e^{-x}) \right]$$

$$z e^{-x} = x e^{-x} + e^{-x}$$

$$\therefore z = x + 1$$

$$\therefore y^{-1} = x + 1$$

$$\therefore y = \frac{1}{1+x}$$

$$\text{②} \quad x \cdot \frac{dy}{dx} - y = x^2 y^3$$

⇒ Divide with  $x$ ,

$$\frac{dy}{dx} - \frac{y}{x} = x y^3 \quad \text{--- ①}$$

$$\textcircled{1} \div \textcircled{1} \text{ with } y^3, \\ y^{-3} \frac{dy}{dx} - y^{-2} x^{-1} = x \quad \text{--- } \textcircled{2}$$

Substitute  $z = y^{-2}$

$$\frac{dz}{dx} = -2y^{-3} \frac{dy}{dx}$$

$$-\frac{1}{2} \frac{dz}{dx} = y^{-3} \frac{dy}{dx}$$

$$\textcircled{2} \Rightarrow -\frac{1}{2} \frac{dz}{dx} - x^{-1} z = x$$

multiply by  $-2$ ,

$$\frac{dz}{dx} + \frac{2z}{x} = -2x \quad \text{--- } \textcircled{3}$$

$\therefore \textcircled{3}$  is linear.

$$P(x) = \frac{2}{x}, \quad Q(x) = -2x$$

$$\therefore \text{I.F.} = e^{\int P(x) dx} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} \\ \text{I.F.} = x^2$$

$$z \cdot \text{I.F.} = \int \text{I.F.} \cdot Q(x) dx + C$$

$$zx^2 = \int x^2 \cdot (-2x) dx + C$$

$$zx^2 = -2 \int x^3 dx + C$$

$$zx^2 = -2 \left( \frac{x^4}{4} \right) + C$$

$$zx^2 = -\frac{x^4}{2} + C$$

$\therefore$  Divide by  $x^2$ ,

$$z = -\frac{x^2}{2} + Cx^{-2}$$

$$\therefore y^{-2} = -\frac{x^2}{2} + \frac{C}{x^2}$$

$\Rightarrow$  Orthogonal Trajectories:-

Two family of curves are orthogonal trajectories of each other, if every member of one family cuts every member of the other family at right angles.

$\Rightarrow$  Methods of Cartesian co-ordinates.

If a family of curves is given in the form  $F(x, y, c) = 0$ .

Find the differential <sup>equation</sup> ~~coordinates~~ of the ~~para~~ family by eliminating  $c$ , i.e.,  $f(x, y, \frac{dy}{dx}) = 0$ .

For orthogonal trajectories replace  $\frac{dy}{dx}$  by  $-\frac{dx}{dy}$ . i.e.,  $f(x, y, -\frac{dx}{dy}) = 0$ .

Simplify to get the D.E. of orthogonal trajectories - solve the D.E.

① Find the orthogonal trajectories of family of parabolas.  $y^2 = 4ax$ .

$\Rightarrow y^2 = 4ax$  --- ①  
diff. ① wrt  $x$ ,

$$\Rightarrow 2y \frac{dy}{dx} = 4a$$

$$\Rightarrow a = \frac{1}{2} y \frac{dy}{dx}$$

$$\textcircled{1} \Rightarrow y^2 = 4 \left( \frac{1}{2} y \frac{dy}{dx} \right) x$$

$$y^2 = 2xy \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{2x}$$

replace  $\frac{dy}{dx}$  with  $-\frac{dx}{dy}$ ,

$$\Rightarrow -\frac{dx}{dy} = \frac{y}{2x}$$

$$\Rightarrow -2x dx = y dy$$

Solving by variable separable method,

$$\Rightarrow \int -2x dx = \int y dy$$

$$\Rightarrow -x^2 = \frac{y^2}{2} + C$$

$$\Rightarrow x^2 + \frac{y^2}{2} = C \rightarrow \text{Ellipse.}$$

$\Rightarrow$  To find orthogonal trajectories in polar co-ordinates, the family is given in the form  $F(r, \theta, C) = 0$ . For orthogonal trajectories replace  $\frac{dr}{d\theta}$  by  $-\frac{r^2}{dr/d\theta}$

① Find the orthogonal trajectories of families of curves.  $r = a(1 + \cos \theta)$

$$\Rightarrow a = \frac{r}{1 + \cos \theta}$$

$$\frac{dr}{d\theta} = a(-\sin \theta)$$

$$\Rightarrow \frac{dr}{d\theta} = \frac{-r \sin \theta}{1 + \cos \theta}$$

replace  $\frac{dr}{d\theta}$  by  $-\frac{r^2}{dr/d\theta}$ ,

$$\Rightarrow r^2 \frac{d\theta}{dr} = \frac{-r \sin \theta}{1 + \cos \theta}$$

$$\Rightarrow r \frac{d\theta}{dr} = \left( \frac{\sin \theta}{1 + \cos \theta} \right)$$

$$\Rightarrow \left( \frac{1 + \cos \theta}{\sin \theta} \right) d\theta = \frac{dr}{r}$$

$$\text{Solving } \int \frac{dr}{r} = \int \frac{1}{\sin \theta} d\theta + \int \frac{\cos \theta}{\sin \theta} d\theta$$

$$\Rightarrow \ln r = \ln(\sec \theta - \cot \theta) + \ln(\sin \theta)$$

$$\Rightarrow \ln r = \ln[(\sec \theta - \cot \theta) \sin \theta]$$

$$\ln r = \ln[1 - \cos \theta] + C$$

$$\therefore r = C(1 - \cos \theta)$$



$\rightarrow$  Solve the following differential equations:-

①  $\underbrace{(y \cos x + 2xe^x)}_M dx + \underbrace{(\sin x + x^2 e^x - 2y)}_N dy = 0$

②  $(xy \sin xy + \cos xy)y dx + x(xy \sin xy + \cos xy) dy = 0$

③  $\frac{dy}{dx} + y \sin x = e^{-\cos x} y^3$

$\rightarrow$  Find the orthogonal trajectory of the following family of curves.

④  $x^2 + y^2 = 2C$

⑤  $x + y = 2$

$\Rightarrow$  ⑥ Exact D.E:-

$$\frac{\partial M}{\partial y} = \cos x + 2xe^x$$

$$\frac{\partial N}{\partial x} = \cos x + 2xe^x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{Exact}$$

Sol<sup>n</sup>:-

$$\int_{y \text{ const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\int (y \cos x + 2xe^y) dx + \int -dy dy = C$$

$$y \sin x + \frac{2x^2}{2} e^y - \frac{dy^2}{2} = C$$

$$\Rightarrow y \sin x + x^2 e^y - y^2 = C$$

$$\Rightarrow (xy \sin xy + \cos xy) y dx + x(xy \sin xy + \cos xy) dy = 0$$

$$\Rightarrow (xy \sin xy + \cos xy) (y dx + x dy) = 0$$

$$\text{Take } u = xy,$$

$$du = y dx + x dy$$

$$\Rightarrow \int (u \sin u + \cos u) du = 0$$

$$-u \cos u + \sin u + \sin u = 0$$

$$-u \cos u + 2 \sin u = C$$

$$-xy \cos(xy) + 2 \sin(xy) = C$$

$$\Rightarrow 2 \sin(xy) - xy \cos(xy) = C$$

$$\textcircled{a} (xy \sin xy + \cos xy) y dx + x(xy \sin xy + \cos xy) dy = 0$$

$\Rightarrow$  Solvable to exact form:-

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\Rightarrow (xy^2 \sin xy + y \cos xy) \frac{dx}{dy} + (x^2 y \sin xy + x \cos xy) \frac{dy}{dx} = 0$$

M

N

$$\frac{\partial M}{\partial y} = 2xy \sin(xy) + xy^2 [x \cos(xy)] + \cos(xy) + y[-\sin(xy) \cdot x]$$

$$\frac{\partial M}{\partial y} = 2xy \sin(xy) + x^2 y^2 \cos(xy) + \cos(xy) - x^2 y \sin(xy)$$

$$\Rightarrow \frac{\partial M}{\partial y} = xy \sin(xy) + x^2 y^2 \cos(xy) + \cos(xy)$$

$$\frac{\partial N}{\partial x} = 2xy \sin(xy) + x^2 y [y \cos(xy)] + \cos(xy) + x[-\sin(xy) \cdot y]$$

$$\frac{\partial N}{\partial x} = 2xy \sin(xy) + x^2 y^2 \cos(xy) + \cos(xy) - xy \sin(xy)$$

$$\Rightarrow \frac{\partial N}{\partial x} = xy \sin(xy) + x^2 y^2 \cos(xy) + \cos(xy)$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{Exact}$$

Sol<sup>n</sup>:-

$$\int_{y \text{ const}} M dx + \int_{\text{terms without } x} N dy = C$$

$$\text{Take } u = xy \Rightarrow du = y dx$$

$$\int y(xy \sin xy + \cos xy) dx + \int 0 dy = C$$

$$\text{Take } u = xy \Rightarrow du = y dx \text{ (since } y \text{ is const)}$$

$$\int (u \sin u + \cos u) du = C$$

$$\Rightarrow -u \cos u + \sin u + \sin u = C$$

$$\Rightarrow -u \cos u + 2 \sin u = C$$

$$\Rightarrow 2 \sin(xy) - xy \cos(xy) = C$$

$$(3) \frac{dy}{dx} + y \sin x = e^{-\cos x} y^3 \quad \text{--- (1)}$$

⇒ Bernoulli's DE:-

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Here,  $P(x) = \sin x$ ,  $Q(x) = e^{-\cos x}$

Divide (1) with  $y^3$ ,

$$\Rightarrow y^{-3} \frac{dy}{dx} + y^{-2} \sin x = e^{-\cos x} \quad \text{--- (2)}$$

Substitute

$$z = y^{-2}$$

$$\frac{dz}{dx} = -2y^{-3} \frac{dy}{dx}$$

$$(2) \Rightarrow -\frac{1}{2} \frac{dz}{dx} + z \sin x = e^{-\cos x}$$

Multiply by  $-2$ ,

$$\frac{dz}{dx} - 2z \sin x = -2e^{-\cos x} \quad \text{--- (3)}$$

∴ (3) is linear,

$$P(x) = -2 \sin x, \quad Q(x) = -2e^{-\cos x}$$

$$\begin{aligned} \text{I.F.} &= e^{\int P(x) dx} = e^{\int -2 \sin x dx} \\ &= e^{2 \cos x} \end{aligned}$$

$$\therefore z \cdot \text{I.F.} = \int \text{I.F.} \cdot Q(x) dx + C$$

$$ze^{2 \cos x} = \int e^{2 \cos x} \cdot -2e^{-\cos x} dx + C$$

$$ze^{2 \cos x} = -2e^{2 \cos x - \cos x}$$

$$ze^{2 \cos x} = -2e^{\cos x}$$

$$y^{-2} = \frac{-2e^{\cos x}}{e^{2 \cos x}} \Rightarrow y^{-2} = -2e^{\cos x - 2 \cos x}$$

$$\Rightarrow y^{-2} = -2e^{-\cos x}$$

~~$$y^{-2} = -2e^{-\cos x}$$~~

$$(4) x^2 + y^2 = 2c \quad \text{--- (1)}$$

⇒ diff (1) wrt  $x$ .

$$2x + 2y \frac{dy}{dx} = 0$$

Divide by 2,

$$x + y \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{x}{y}$$

replace  $\frac{dy}{dx}$  with  $-\frac{dx}{dy}$

$$\Rightarrow -\frac{dx}{dy} = -\frac{x}{y}$$

$$\Rightarrow \frac{1}{x} dx = \frac{1}{y} dy$$

Solving with variable separable,

$$\Rightarrow \int \frac{1}{x} dx = \int \frac{1}{y} dy$$

$$\Rightarrow \log x = \log y + \log C$$

$$\Rightarrow \log x - \log y = \log C$$

$$\Rightarrow \log \frac{x}{y} = \log C$$

$$\Rightarrow \frac{x}{y} = C$$

$$\textcircled{5} \quad x + y = 2 \quad - \textcircled{1}$$

$\Rightarrow$  diff  $\textcircled{1}$  wrt  $x$ ,

$$1 + \frac{dy}{dx} = 0.$$

$$\frac{dy}{dx} = -1$$

replace  $\frac{dy}{dx}$  with  $-\frac{dx}{dy}$

$$\Rightarrow -\frac{dx}{dy} = -1$$

$$\Rightarrow \int dx = \int dy. \Rightarrow \underline{\underline{x = y + c}}$$